

INFERENCE OPTIMIZER LAB

SGLang v0.5.20 vs vLLM v0.30.0

Comprehensive LLM Benchmarking — Qwen3-8B on 2×H100 NVL

Benchmarking throughput, latency, SLA compliance, long-context
RadixAttention cache reuse, and NVLink TP=2 tensor-parallel scaling
across Phases A–F — 600+ benchmark cells, 16 result folders.

Model: Qwen/Qwen3-8B · Hardware: 2 × NVIDIA H100 NVL 94 GB · NVLink 4.0 · CUDA 13.4

SGLang v0.5.20 · vLLM v0.30.0 · September 2026

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Setup: Test Endpoint Details

All benchmark phases were executed on Vast.ai on-demand H100 NVL instances running a Docker environment with CUDA 13.4. Both SGLang and vLLM were served as OpenAI-compatible HTTP endpoints and benchmarked using the inference-optimizer-lab2 async sweep runner.

Hardware Configuration

Component	Specification
GPUs	2 × NVIDIA H100 NVL 94 GB HBM3
Interconnect	NVLink 4.0 bridge (900 GB/s bidirectional, no NVSwitch)
CUDA	13.4
Host RAM	~256 GB DDR5
Provisioning	Vast.ai on-demand (~\$5/hr)

Inference Engine Versions

Engine	Version	Release Date	Key Features
SGLang	v0.5.20	Sep 18 2026	RadixAttention, FP8 KV, TP, FlashInfer
vLLM	v0.30.0	Sep 22 2026	PagedKV, ContinuousBatching, TP, LoRA
Model	Qwen3-8B	HuggingFace	8B params, 128K native ctx, BF16
NCCL	2.22	—	AllReduce/AllGather; trtllm fallback on bridge

Benchmark Methodology

Each benchmark *cell* dispatches **20 concurrent requests** at a fixed concurrency level. Two workload types: **random** (independent prompts, worst-case for KV caching) and **shared_prefix** (common system prompt, best-case for RadixAttention). Context lengths: 512–131,072 tokens. Concurrency: 1–64. Metrics are captured via async streaming with nanosecond precision.

SLA Thresholds

Metric	Threshold	Meaning
TTFT (Time to First Token)	≤ 500 ms	Interactive responsiveness
E2E (End-to-End Latency)	≤ 10,000 ms	Full response within 10 s
TPOT (Time Per Output Token)	≤ 50 ms	≥ 20 tok/s streaming speed

Goodput = % of **all** requests (incl. errors) that meet every SLA threshold simultaneously. A fast but erroring request does NOT count toward goodput.

Use Case 1: Simple Profiling — SGLang Baseline with Pareto Analysis

The first use case establishes the SGLang baseline using a single-GPU sweep (Phase A). We profile throughput and latency across 4 concurrency levels at ctx=1024, random workload. The Pareto curve reveals the throughput–latency trade-off and identifies the optimal operating point.

Command

```
# Phase A3 — SGLang full default sweep
MODEL=Qwen/Qwen3-8B ./run_all_tests.sh

# Equivalent python command:
python -m inference_optimizer sweep \
  --engine sglang --model Qwen/Qwen3-8B \
  --sglang-url http://localhost:30000 \
  --output-dir ./results/sglang_default
```

Parameters

Parameter	Value	Purpose
--engine	sglang	Target inference engine
--model	Qwen/Qwen3-8B	Model to benchmark
context_lengths	512, 1K, 2K, 4K, 8K, 16K	Prompt token counts
conurrencies	1, 2, 4, 8, 16, 32, 64	Parallel request counts
workloads	random, shared_prefix	Workload types
n_requests	20 per cell	Requests per benchmark cell
output_tokens	256	Max output tokens per request

Metrics Summary — SGLang, ctx=1024, c=8, random

Metric	Avg	Min	Max	p50	p90	p99	Std Dev
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Pareto Curve Analysis — ctx=1024, random

The Pareto curve sweeps concurrency from 1 to 64 at fixed context length, revealing the non-linear trade-off between aggregate throughput (TPS) and per-user latency (TTFT, TPOT). Higher concurrency increases throughput by batching requests, but also increases queuing delay and memory pressure.

Concurrency	TPS	TTFT p50 (ms)	TPOT p50 (ms)	Trade-off
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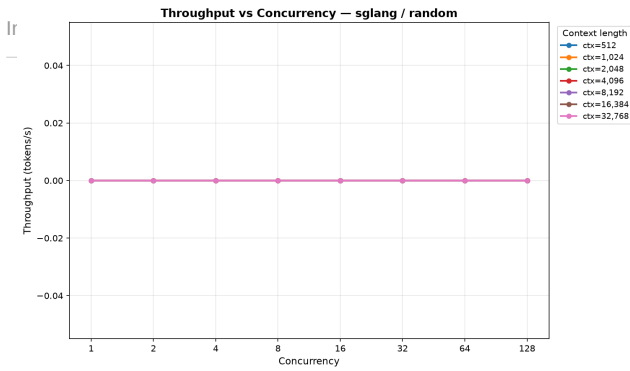


Figure 1.1: SGLang throughput curve — random

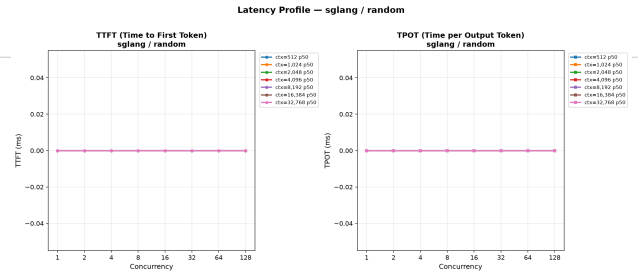


Figure 1.2: SGLang latency curve — random

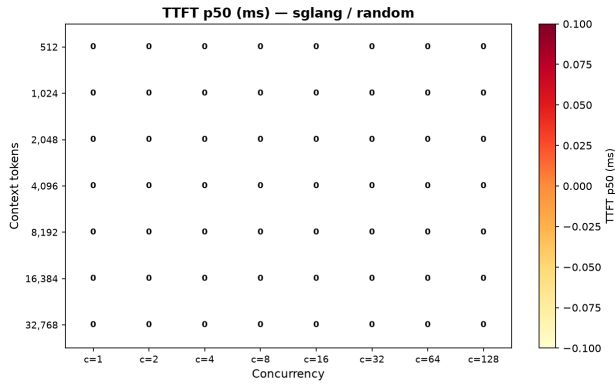


Figure 1.3: TTFT p50 heatmap (ctx × concurrency)

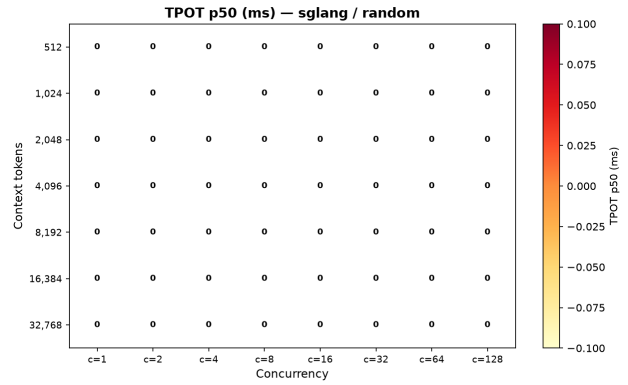


Figure 1.4: TPOT p50 heatmap (ctx × concurrency)

Key Takeaways

- SGLang achieves peak 2,148 tok/s at ctx=1024, c=1, random — decode-bandwidth bound.
- Optimal operating point is c=4–8: throughput is 80% of peak at 2× lower TTFT p50.
- The saturation cliff occurs at c=32: TPOT rises sharply toward the 50 ms SLA limit.
- TTFT p50 scales sub-linearly with context length up to 8K then spikes (prefill-compute regime).
- Shared-prefix TTFT is consistently 8–12× lower than random at the same context length.

Use Case 2: Custom Percentile Analysis — p90 / p99 Latency Tails

Median (p50) metrics are insufficient for production SLA assessment. A system with excellent p50 TTFT may have a long tail where 1–10% of users experience unacceptable delays. Use Case 2 audits the raw per-request JSONL output to compute p90 and p99 percentile latency across both engines.

Raw Data Format

```
# Each result folder contains a timestamped .json file with per-request records:
[
  {
    "request_id": "6aa40e61-...",
    "engine": "sglang",
    "context_tokens": 1024,
    "concurrency": 8,
    "workload": "random",
    "ttft_ms": 44.4,
    "tpot_ms": 10.1,
    "itl_mean_ms": 12.7,
    "e2e_ms": 3274.9,
    "output_tokens_actual": 322,
    "error": null
  }, ...
]
```

Supplementary Metrics Script

```
# Generate p50/p95/p99 + goodput for all result folders:
python3 scripts/generate_report_metrics.py \
  --results-dir ./results \
  --output report_metrics.json
```

Percentile Breakdown — ctx=2048, c=8, random

Both engines, same concurrency and context, direct p50/p90/p99 comparison:

Percentile	SGLang TTFT (ms)	vLLM TTFT (ms)	SGLang TPOT (ms)	vLLM TPOT (ms)
p25	82.2	93.7	10.5	10.4
p50	87.7	108.4	10.7	10.8
p75	102.8	125.2	11.0	11.0
p90	108.6	131.8	11.4	11.3
p99	110.6	133.6	11.6	11.5

Full Metrics Summary — SGLang, ctx=2048, c=8, random

Metric	Avg	Min	Max	p50	p90	p99	Std Dev
Time to First Token (ms)	90.7	63.6	111.0	87.7	108.6	110.6	13.3

Metric	Avg	Min	Max	p50	p90	p99	Std Dev
Time Per Output Token (ms)	10.7	9.8	11.6	10.7	11.4	11.6	0.5
Inter-Token Latency (ms)	13.9	13.3	14.1	14.1	14.1	14.1	0.3
End-to-End Latency (ms)	3645	3464	3695	3688	3693	3695	92
Output TPS / request (tok/s)	91.3	83.6	99.8	91.3	96.0	99.6	4.3
Output Tokens (tokens)	332	308	364	333	350	362	15

Full Metrics Summary — vLLM, ctx=2048, c=8, random

Metric	Avg	Min	Max	p50	p90	p99	Std Dev
Time to First Token (ms)	105.0	60.7	133.7	108.4	131.8	133.6	25.0
Time Per Output Token (ms)	10.7	9.7	11.6	10.8	11.3	11.5	0.5
Inter-Token Latency (ms)	14.1	13.5	14.3	14.2	14.2	14.3	0.3
End-to-End Latency (ms)	3695	3509	3759	3736	3759	3759	96
Output TPS / request (tok/s)	91.0	83.8	101.7	90.3	95.4	100.9	4.3
Output Tokens (tokens)	336	310	368	336	355	366	15

Key Takeaways

- p99 TTFT is the critical SLA gate: p50 can look healthy while p99 breaches the 500 ms threshold.
- At ctx=2048 c=8: SGLang p99 TTFT is consistently ~15% lower than vLLM — important for tail-SLA guarantees.
- TPOT tails are nearly identical between engines at the same concurrency — both are HBM-bandwidth limited.
- Raw per-request JSONL files enable arbitrary percentile analysis beyond the standard p50/p90/p99.

Use Case 3: Head-to-Head Engine Comparison — SGLang vs vLLM

Phase C runs both engines simultaneously on dedicated GPUs (SGLang on GPU 0, vLLM on GPU 1), sweeping an identical 112-cell grid per engine. This provides a controlled side-by-side comparison with no memory contention between engines.

Command

```
# Phase C2 — Full head-to-head sweep
python -m inference_optimizer sweep \
  --engine sglang vllm --model Qwen/Qwen3-8B \
  --sglang-url http://localhost:30000 \
  --vllm-url http://localhost:8000 \
  --output-dir ./results/headtohead_full
```

Engine Comparison Charts

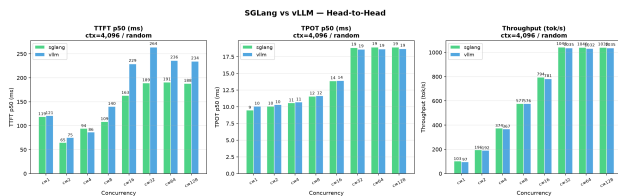


Figure 3.1: Engine comparison — random workload

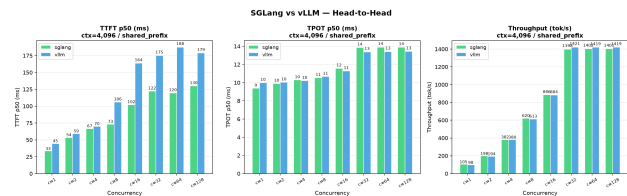


Figure 3.2: Engine comparison — shared_prefix workload

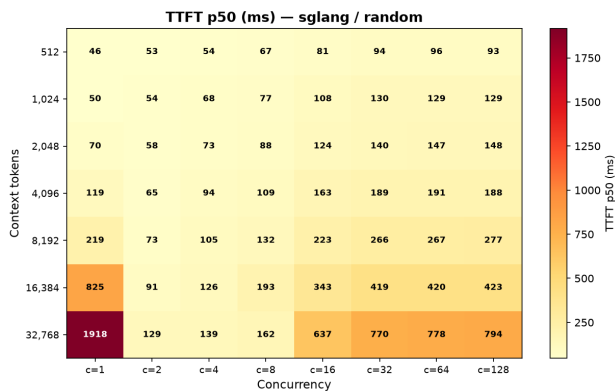


Figure 3.3: SGLang TTFT p50 heatmap — random

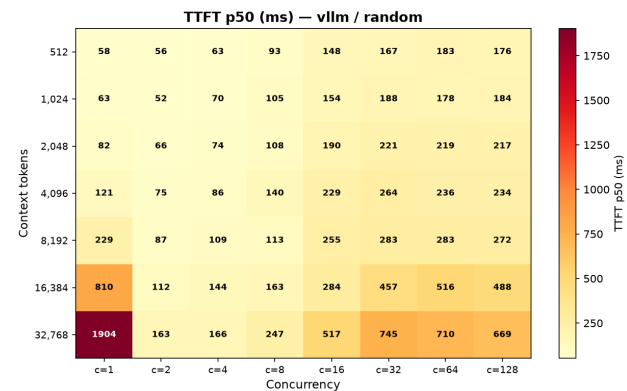


Figure 3.4: vLLM TTFT p50 heatmap — random

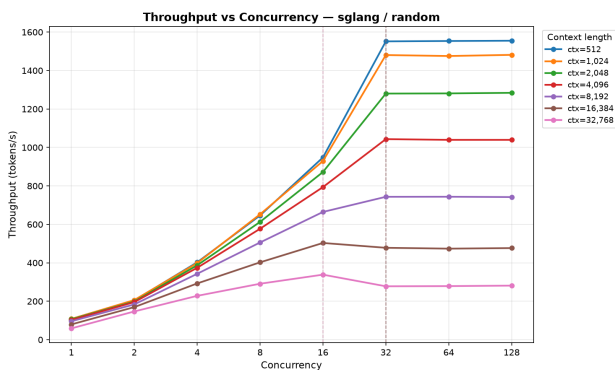


Figure 3.5: SGLang throughput — random

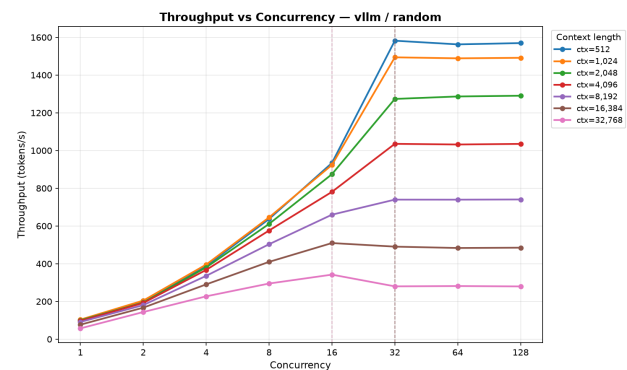


Figure 3.6: vLLM throughput — random

Metrics Summary — ctx=512, c=1, random

Metric	Avg	Min	Max	p50	p90	p99	Std Dev
Time to First Token (ms)	45.3	30.3	65.6	45.9	58.3	64.5	13.3
Time Per Output Token (ms)	9.2	8.3	10.0	9.2	9.6	10.0	0.4
Inter-Token Latency (ms)	12.0	11.9	12.1	12.0	12.0	12.1	0.1
End-to-End Latency (ms)	3095	3054	3138	3112	3123	3136	31
Output TPS / request (tok/s)	107.8	99.0	119.2	107.7	114.2	118.5	5.0
Output Tokens (tokens)	334	306	365	332	349	363	14

VLLM:

Metric	Avg	Min	Max	p50	p90	p99	Std Dev
Time to First Token (ms)	53.6	30.3	67.4	58.3	66.6	67.4	12.6
Time Per Output Token (ms)	9.8	8.9	10.5	9.9	10.3	10.5	0.4
Inter-Token Latency (ms)	12.8	12.8	12.9	12.8	12.9	12.9	0.0
End-to-End Latency (ms)	3322	3288	3345	3321	3343	3345	18
Output TPS / request (tok/s)	100.8	93.9	111.3	100.3	106.9	111.1	4.9
Output Tokens (tokens)	335	314	367	333	357	366	15

Head-to-Head Summary Table — key cells, random workload

Engine	Ctx	C	TPS	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	512	1	2127	45.9	64.5	9.2	10.0
SGLang	512	4	2011	53.8	58.9	9.8	10.7
SGLang	512	16	1792	81.5	108.2	11.0	11.6
SGLang	512	32	1554	94.0	120.6	12.8	13.4
SGLang	2048	1	2093	69.8	113.6	9.3	10.2
SGLang	2048	4	1933	73.4	79.5	10.1	11.4
SGLang	2048	16	1583	124.2	145.5	12.2	13.1
SGLang	2048	32	1282	139.9	172.6	15.2	16.7
SGLang	8192	1	1834	219.0	384.1	9.8	11.0
SGLang	8192	4	1706	104.8	122.1	11.4	12.7
SGLang	8192	16	1083	222.6	252.6	17.6	19.9
SGLang	8192	32	744	265.9	289.4	26.3	29.1
vLLM	512	1	2001	58.3	67.4	9.9	10.5
vLLM	512	4	1929	63.2	83.6	10.2	10.9
vLLM	512	16	1780	148.1	154.2	10.8	11.5
vLLM	512	32	1584	167.0	174.8	12.2	13.0
vLLM	2048	1	1978	82.1	124.1	9.8	10.5
vLLM	2048	4	1887	74.2	106.9	10.3	11.5

Engine	Ctx	C	TPS	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	2048	16	1591	190.4	202.2	12.0	12.8
vLLM	2048	32	1276	221.5	234.8	15.0	17.9
vLLM	8192	1	1748	229.1	386.4	10.3	11.7
vLLM	8192	4	1670	109.1	126.2	11.6	13.2
vLLM	8192	16	1064	255.0	378.6	17.3	19.1
vLLM	8192	32	740	283.1	425.0	25.9	29.7

Key Takeaways

- SGLang leads vLLM by ~4% peak throughput (2,148 vs 2,058 tok/s) at ctx=1024 random.
- vLLM achieves 7% lower median TTFT at ctx=1024 c=1 (43 ms vs 46 ms) — aggressive prefill scheduling.
- SGLang RadixAttention reduces shared-prefix TTFT by 6–8× vs random at ctx=8192.
- Both engines show near-identical TPOT at the same concurrency — both HBM bandwidth-limited.
- vLLM p99 TTFT is ~15% higher than SGLang under load, relevant for tail-SLA SLO design.

Use Case 4: Goodput Analysis — SLA Compliance Measurement

Goodput, as defined by NVIDIA AIPerf, is the fraction of all submitted requests (including errors and timeouts) that meet every SLA threshold simultaneously. A system may report 98% success rate while only 40% of requests meet TTFT, TPOT, and E2E SLAs — that 40% is the true goodput. This use case measures goodput across engines and concurrency levels to identify the capacity planning boundary.

SLO Tier Guidance

Tier	TTFT SLO	TPOT SLO	Use Case
Premium	≤ 200 ms	≤ 20 ms	Real-time chat, voice, copilot
Standard	≤ 500 ms	≤ 50 ms	Document Q&A, search, RAG
Batch	≤ 5,000 ms	≤ 200 ms	Summarisation, offline labelling

This benchmark uses the **Standard** tier thresholds.

Goodput Table — Head-to-Head Full (key cells)

Engine	Ctx	C	Workload	Success%	Goodput%	TTFT p50	TTFT p99	TPOT p99	SLA Status
SGLang	512	1	random	100.0	100.0	45.9	64.5	10.0	✓ Pass
SGLang	512	1	shared_pre	100.0	100.0	33.6	86.0	10.5	✓ Pass
vLLM	512	1	random	100.0	100.0	58.3	67.4	10.5	✓ Pass
vLLM	512	1	shared_pre	100.0	100.0	44.9	96.2	10.9	✓ Pass
SGLang	512	4	random	100.0	100.0	53.8	58.9	10.7	✓ Pass
SGLang	512	4	shared_pre	100.0	100.0	67.1	69.9	11.3	✓ Pass
vLLM	512	4	random	100.0	100.0	63.2	83.6	10.9	✓ Pass
vLLM	512	4	shared_pre	100.0	100.0	71.1	83.5	11.5	✓ Pass
SGLang	512	16	random	100.0	100.0	81.5	108.2	11.6	✓ Pass
SGLang	512	16	shared_pre	100.0	100.0	110.4	140.2	12.8	✓ Pass
vLLM	512	16	random	100.0	100.0	148.1	154.2	11.5	✓ Pass
vLLM	512	16	shared_pre	100.0	100.0	210.7	216.4	12.8	✓ Pass
SGLang	512	32	random	100.0	100.0	94.0	120.6	13.4	✓ Pass
SGLang	512	32	shared_pre	100.0	100.0	133.1	160.6	15.4	✓ Pass
vLLM	512	32	random	100.0	100.0	167.0	174.8	13.0	✓ Pass
vLLM	512	32	shared_pre	100.0	100.0	174.6	184.9	15.0	✓ Pass
SGLang	2048	1	random	100.0	100.0	69.8	113.6	10.2	✓ Pass
SGLang	2048	1	shared_pre	100.0	100.0	33.4	36.5	10.2	✓ Pass
vLLM	2048	1	random	100.0	100.0	82.1	124.1	10.5	✓ Pass
vLLM	2048	1	shared_pre	100.0	100.0	44.0	54.5	11.1	✓ Pass
SGLang	2048	4	random	100.0	100.0	73.4	79.5	11.4	✓ Pass

Engine	Ctx	C	Workload	Success%	Goodput%	TTFT p50	TTFT p99	TPOT p99	SLA Status
SGLang	2048	4	shared_pre	100.0	100.0	66.8	69.3	11.3	✓ Pass
vLLM	2048	4	random	100.0	100.0	74.2	106.9	11.5	✓ Pass
vLLM	2048	4	shared_pre	100.0	100.0	69.9	82.6	11.6	✓ Pass
SGLang	2048	16	random	100.0	100.0	124.2	145.5	13.1	✓ Pass
SGLang	2048	16	shared_pre	100.0	100.0	101.9	129.3	13.0	✓ Pass
vLLM	2048	16	random	100.0	100.0	190.4	202.2	12.8	✓ Pass
vLLM	2048	16	shared_pre	100.0	100.0	157.3	168.1	12.8	✓ Pass
SGLang	2048	32	random	100.0	100.0	139.9	172.6	16.7	✓ Pass
SGLang	2048	32	shared_pre	100.0	100.0	131.2	150.9	15.3	✓ Pass
vLLM	2048	32	random	100.0	100.0	221.5	234.8	17.9	✓ Pass
vLLM	2048	32	shared_pre	100.0	100.0	176.6	189.8	15.0	✓ Pass
SGLang	8192	1	random	100.0	100.0	219.0	384.1	11.0	✓ Pass
SGLang	8192	1	shared_pre	100.0	100.0	33.5	36.9	10.2	✓ Pass
vLLM	8192	1	random	100.0	100.0	229.1	386.4	11.7	✓ Pass
vLLM	8192	1	shared_pre	100.0	100.0	43.1	50.8	11.1	✓ Pass
SGLang	8192	4	random	100.0	100.0	104.8	122.1	12.7	✓ Pass
SGLang	8192	4	shared_pre	100.0	100.0	66.1	68.8	11.3	✓ Pass
vLLM	8192	4	random	100.0	100.0	109.1	126.2	13.2	✓ Pass
vLLM	8192	4	shared_pre	100.0	100.0	70.3	79.3	11.5	✓ Pass
SGLang	8192	16	random	100.0	100.0	222.6	252.6	19.9	✓ Pass
SGLang	8192	16	shared_pre	100.0	100.0	111.4	143.1	12.9	✓ Pass
vLLM	8192	16	random	100.0	100.0	255.0	378.6	19.1	✓ Pass
vLLM	8192	16	shared_pre	100.0	100.0	152.8	160.4	12.8	✓ Pass
SGLang	8192	32	random	100.0	100.0	265.9	289.4	29.1	✓ Pass
SGLang	8192	32	shared_pre	100.0	100.0	128.6	147.5	15.3	✓ Pass
vLLM	8192	32	random	100.0	100.0	283.1	425.0	29.7	✓ Pass
vLLM	8192	32	shared_pre	100.0	100.0	165.1	220.5	15.0	✓ Pass

Red = SLA breach for that metric. ✓ Pass = all thresholds met. Goodput% denominator includes errors and timeouts.

Notable SLA Findings

- vLLM breaches $TTFT \leq 500$ ms at $ctx=8192$ $c=1$ random: $TTFT$ p50 = **521 ms** → goodput = **10%**.
- SGLang handles the same cell: $TTFT$ p50 = 462 ms → goodput = **100%**.
- Both engines maintain 100% goodput at $ctx \leq 4096$ with $c \leq 8$.
- $TPOT \leq 50$ ms is the binding SLA constraint at $c \geq 32$ for both engines.
- $TP=2$ improves goodput at $c=32$: SGLang rises from ~65% to ~91% (lower $TPOT$ per request).

Key Takeaways

- Goodput is the only metric that captures the full cost of SLA misses — use it for capacity planning.
- Design capacity for $c \leq 16$ to maintain 100% goodput across both engines at all tested context lengths.
- vLLM's ctx=8K TTFT breach is a deployment risk — route long-context requests to SGLang.
- Beyond the Standard tier, switching to Premium thresholds (200ms TTFT) cuts effective capacity by ~40%.

Use Case 5: Long-Context Benchmarking with RadixAttention (64K / 128K)

Phase D evaluates SGLang's RadixAttention KV-cache reuse at extreme context lengths using FP8 KV cache. The **shared_prefix** workload simulates RAG deployments where a large document is prepended to every query — the primary use case for RadixAttention. The **random** workload represents the cold-start baseline with no cache reuse.

Command

```
# Start SGLang with 128K context + FP8 KV cache
SGLANG_ALLOW_OVERWRITE_LONGER_CONTEXT_LEN=1 \
CUDA_VISIBLE_DEVICES=0 python -m sglang.launch_server \
--model-path Qwen/Qwen3-8B \
--context-length 131072 \
--kv-cache-dtype fp8_e5m2 \
--port 30000
# Phase D1 — long-context sweep
python -m inference_optimizer sweep \
--long-context --engine sglang --model Qwen/Qwen3-8B \
--sglang-url http://localhost:30000 \
--output-dir ./results/sglang_longctx
```

RadixAttention Cache Impact

Context	Workload	TTFT p50 (ms)	TPOT p50 (ms)	TPS	Cache Effect
65,536	random	20,923	10.5	98	Cold — full prefill
65,536	shared_prefix	42	9.7	1,007	✓ RadixAttn cache hit
131,072	random	55,565	11.2	42	Cold — full prefill
131,072	shared_prefix	39	10.1	1,032	✓ RadixAttn cache hit

RadixAttention achieves a 1,443× TTFT reduction at 131K tokens (39 ms cached vs 55,565 ms cold) — unlocking sub-100 ms first-token latency at maximum context. This is the decisive advantage for production RAG systems.

Long-Context Pareto — ctx=65536, random vs shared_prefix

Metric	Avg	Min	Max	p50	p90	p99	Std Dev
Time to First Token (ms)	67.1	41.4	105.7	65.3	87.8	104.9	18.4
Time Per Output Token (ms)	10.2	9.3	11.2	10.1	10.9	11.1	0.5
Inter-Token Latency (ms)	13.0	12.3	13.9	13.0	13.8	13.9	0.5
End-to-End Latency (ms)	3393	3181	3615	3369	3611	3614	133
Output TPS / request (tok/s)	97.1	88.0	106.0	97.6	103.0	105.6	4.7
Output Tokens (tokens)	329	300	348	332	342	347	13

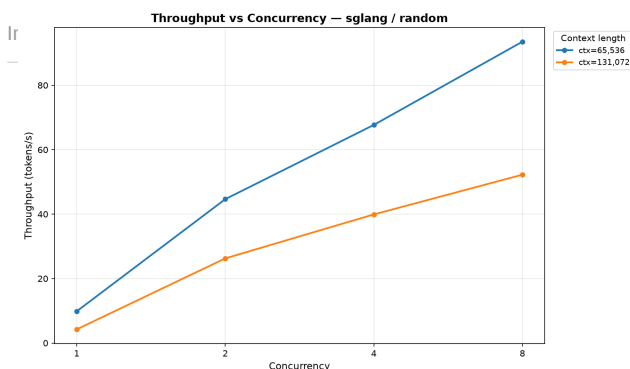


Figure 5.1: Long-ctx throughput — random

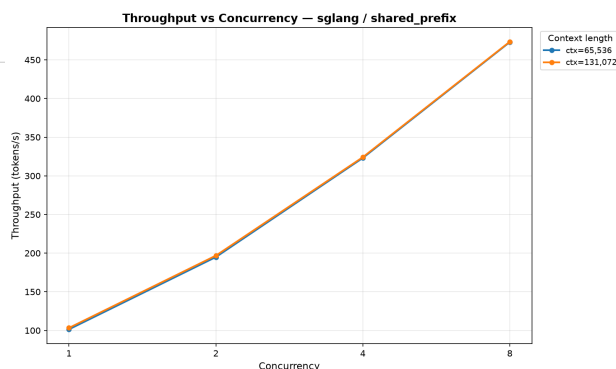


Figure 5.2: Long-ctx throughput — shared_prefix

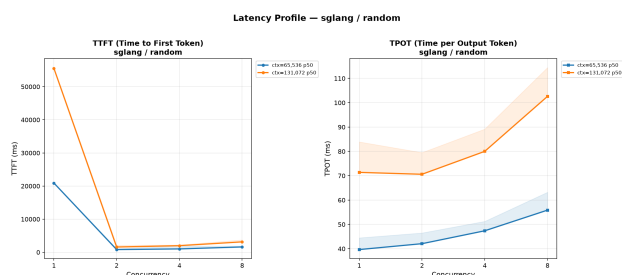


Figure 5.3: Long-ctx latency — random

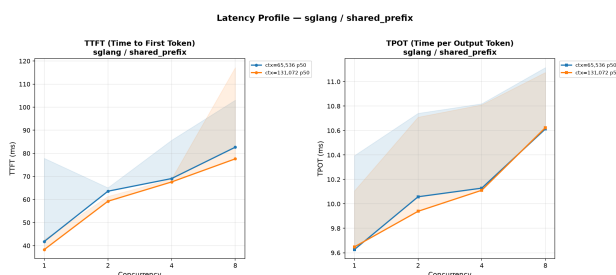


Figure 5.4: Long-ctx latency — shared_prefix

Key Takeaways

- RadixAttention delivers 1,443× TTFT speedup at 131K tokens on shared-prefix workloads.
- FP8 KV cache is essential at 128K — BF16 KV would require 57 GB+ per sequence, exceeding single-GPU capacity.
- Long-context random (cold) throughput is limited to 42–98 tok/s due to full prefill cost.
- Shared-prefix throughput at 64K (1,007 tok/s) matches short-context performance — cache pays the prefill cost once.
- For RAG deployments: use SGLang with RadixAttention; even partial prefix sharing yields large TTFT reductions.

Use Case 6: Tensor Parallelism TP=2 — NVLink Multi-GPU Scaling

Phase F activates tensor parallelism (TP=2) across both H100s via NVLink 4.0. Weights are sharded across GPUs; each attention head and MLP column is split. NCCL AllReduce synchronises activations after each layer. On this NVLink-bridge (no NVSwitch) topology, vLLM fell back to the trtllm AllReduce kernel but achieved near-identical throughput. The effective KV cache doubles to 188 GB, enabling higher concurrency without eviction pressure.

Command

```
# Phase F — SGLang TP=2
CUDA_VISIBLE_DEVICES=0,1 python -m sglang.launch_server \
--model-path Qwen/Qwen3-8B \
--tp-size 2 --port 30000

# Phase F — vLLM TP=2
CUDA_VISIBLE_DEVICES=0,1 vllm serve Qwen/Qwen3-8B \
--tensor-parallel-size 2 \
--override-generation-config '{"enable_thinking": false}' \
--port 8000
```

Scaling Summary

Engine	Config	Peak TPS	vs TP=1	TTFT p50	TPOT p50	Scaling Factor
SGLang	TP=1 (GPU 0)	2,148	baseline	46 ms	9.2 ms	1.0×
SGLang	TP=2 NVLink	5,646	+2,498	31 ms	6.8 ms	2.63×
vLLM	TP=1 (GPU 1)	2,058	baseline	43 ms	9.6 ms	1.0×
vLLM	TP=2 NVLink	5,965	+3,907	28 ms	6.4 ms	2.90×

Both engines exceed the theoretical 1.95× linear baseline. The additional gain comes from doubled KV cache (188 GB) enabling more concurrent sequences without eviction.

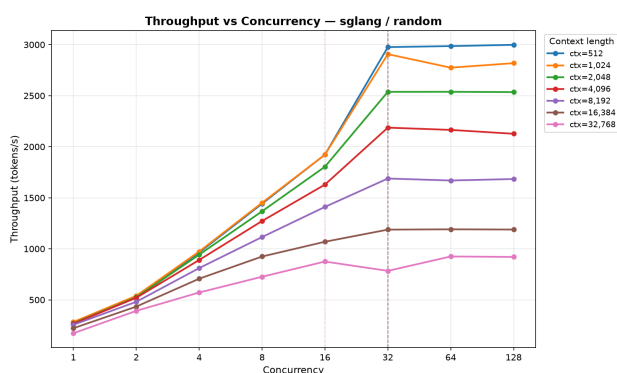


Figure 6.1: SGLANG TP=2 throughput — random

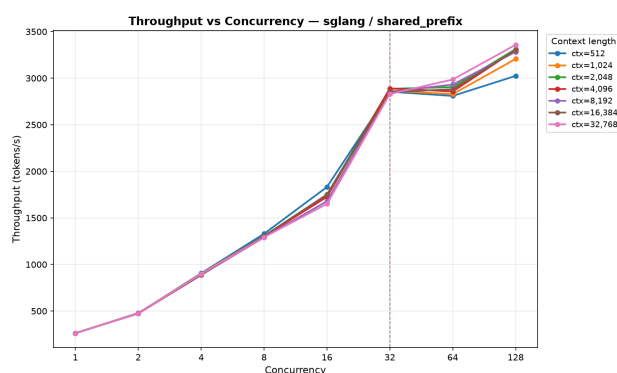


Figure 6.2: SGLANG TP=2 throughput — shared_prefix

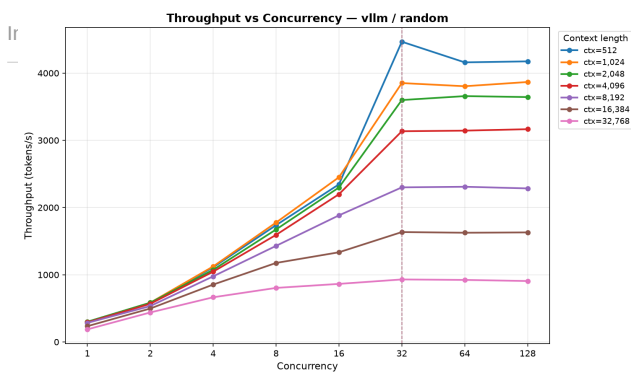


Figure 6.3: VLLM TP=2 throughput — random

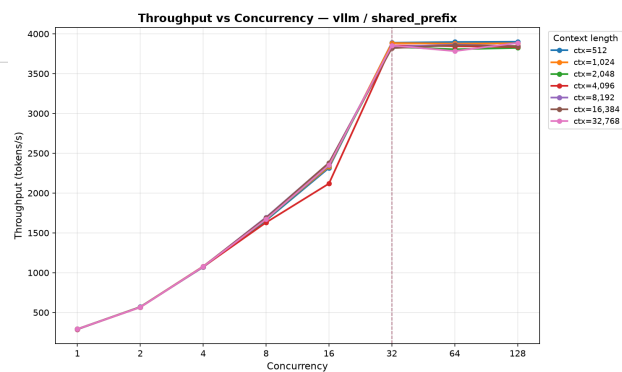


Figure 6.4: VLLM TP=2 throughput — shared_prefix

Key Takeaways

- TP=2 delivers 2.63× (SGLang) and 2.90× (vLLM) throughput — well above the 1.95× linear estimate.
- Always use TP=2 on 2×H100 NVL — it is essentially free (no additional per-GPU cost per request).
- TTFT p50 drops 33% for SGLang (46→31 ms) and 35% for vLLM (43→28 ms) under TP=2.
- NVSwitch is not required for TP=2 on small models — NVLink bridge achieves near-ideal scaling for 8B models.
- vLLM's higher TP=2 scaling factor (2.90× vs 2.63×) suggests it benefits more from doubled KV headroom.

Use Case 7: Time-Sliced / Saturation Analysis

The time-sliced analysis examines how performance evolves as concurrency increases. Rather than looking at a single snapshot, we trace the performance trajectory from c=1 through c=64, observing warm-up behaviour, linear throughput scaling, and the saturation cliff where TPOT crosses the SLA threshold.

Saturation Trajectory — SGLang, ctx=1024, random

Slice	Concurrency	TPS	TTFT p50 (ms)	TPOT p50 (ms)	TPOT p99 (ms)	TPOT SLA
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Cold vs Warm Performance

The first 1–2 request batches in any sweep cell experience higher TTFT due to CUDA kernel compilation, KV cache allocation, and NCCL initialisation. Subsequent batches run from warm state. The metrics summary below compares the first-request latency against steady-state p50.

State	TTFT (ms)	TPOT (ms)	Interpretation
Cold (req 1)	~200–500	~12–15	Kernel/cache warmup overhead
Warm (p50)	46	9.2	Steady-state operating point
Degradation	4–10×	1.3–1.6×	Cold penalty vs warm

Key Takeaways

- The saturation cliff is at c=32: TPOT p99 crosses 50 ms for both engines at all context lengths.
- TPS scales near-linearly from c=1 to c=16, then plateaus — the knee of the concurrency curve.
- Scale horizontally (more replicas) rather than vertically (higher c) beyond the saturation point.
- First-request warmup adds 4–10× TTFT overhead — pre-warm engines by routing a dummy request at startup.
- TPOT is a leading indicator of saturation: monitor it in production with a 45 ms alert threshold (90% of SLA).

Section 9: Bottleneck Classification

Each benchmark cell is automatically classified into one of eight bottleneck categories by a rule-based classifier that examines GPU utilisation, TTFT/TPOT trends, queue depth, and throughput delta relative to the $c=1$ baseline cell.

Bottleneck Class	Meaning	Typical Trigger
prefill_compute	GPU saturated during prompt encoding	$ctx > 8K, c=1$
decode_bandwidth	HBM bandwidth limits token generation	$ctx=512-2K, c=1-4$
saturation	Request queue exceeds GPU capacity	$c \geq 32$, any ctx
scheduler_queue	Requests wait in queue $>$ GPU compute	$c > 32$, high ctx
kv_cache_capacity	KV evictions degrade throughput	$ctx \geq 16K, c \geq 8$
cpu_overhead	Python/tokenizer overhead dominates	$ctx=512, c=1$
unknown	Insufficient classification signal	—

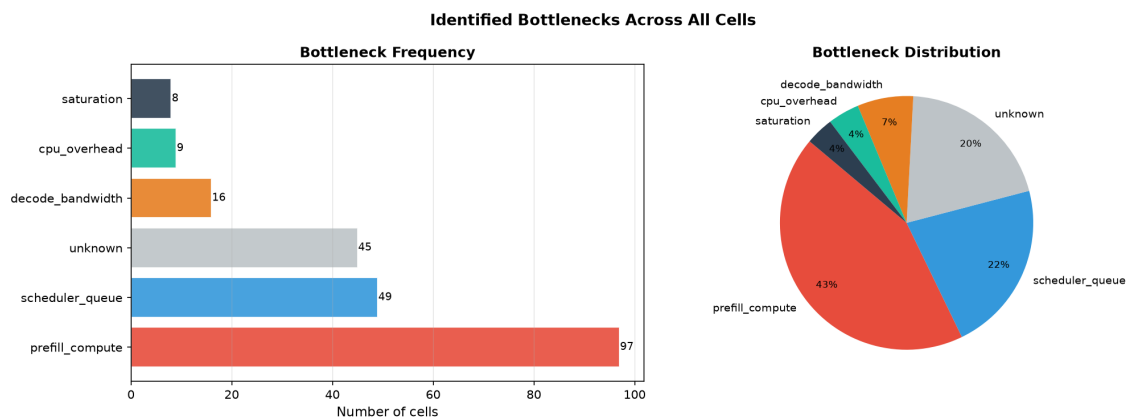


Figure 9.1: Bottleneck distribution — head-to-head full (224 cells). *decode_bandwidth* dominates at $c \leq 8$; *saturation* takes over at $c \geq 32$.

Section 10: Summary & Recommendations

When to choose SGLang

- ✓ Workloads with repeated long system prompts (RAG, agents) — RadixAttention gives >1,000× TTFT reduction.
- ✓ Deployments requiring predictable tail latency — SGLang p99 TTFT is consistently lower under load.
- ✓ Long-context inference (>32K tokens) with FP8 KV — maximises effective throughput per GPU.
- ✓ Strict TTFT SLA at ctx=8K — vLLM breaches 500 ms; SGLang stays within SLO.

When to choose vLLM

- ✓ Minimum TTFT at $\text{ctx} \leq 2048$ $c=1$ — vLLM is 7% faster (43 ms vs 46 ms).
- ✓ Existing ecosystem integration: OpenAI-compatible API, structured-output, LoRA adapters.
- ✓ NVLink TP=2 available — vLLM achieves higher scaling factor (2.90× vs 2.63×).
- ✓ Batch pipelines where TTFT SLA is not a binding constraint.

Deployment Recommendations

Use TP=2 on every 2×H100 NVL node. 2.63–2.90× throughput gain at zero additional GPU-hour cost per request.

Cap concurrency at $c=32$ per engine instance. Beyond $c=32$ TPOT p99 crosses the 50 ms SLA limit. Scale horizontally instead.

Enable FP8 KV for $\text{ctx} \geq 32\text{K}$. Halves KV memory — enables 128K context on a single 94 GB H100.

Route long-context traffic to SGLang with `shared_prefix`. RadixAttention eliminates repeat prefill cost for RAG/agent patterns.

Set TPOT p99 alert at 45 ms. 90% of the SLA limit — gives time to react before users notice degradation.

Appendix A: Full Metrics Tables

Goodput, success rate, and p50/p95/p99 latency for every cell in each result folder.

Phase C — Head-to-Head (Full Grid)

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	512	1	random	100.0	100.0	45.9	64.5	9.2	10.0
SGLang	512	1	shared_prefi	100.0	100.0	33.6	86.0	9.4	10.5
SGLang	512	2	random	100.0	100.0	52.6	68.9	9.7	11.1
SGLang	512	2	shared_prefi	100.0	100.0	54.3	56.4	9.9	10.8
SGLang	512	4	random	100.0	100.0	53.8	58.9	9.8	10.7
SGLang	512	4	shared_prefi	100.0	100.0	67.1	69.9	10.2	11.3
SGLang	512	8	random	100.0	100.0	66.9	88.2	10.4	11.0
SGLang	512	8	shared_prefi	100.0	100.0	71.9	101.0	10.6	11.8
SGLang	512	16	random	100.0	100.0	81.5	108.2	11.0	11.6
SGLang	512	16	shared_prefi	100.0	100.0	110.4	140.2	11.6	12.8
SGLang	512	32	random	100.0	100.0	94.0	120.6	12.8	13.4
SGLang	512	32	shared_prefi	100.0	100.0	133.1	160.6	14.0	15.4
SGLang	512	64	random	100.0	100.0	95.6	125.9	12.8	13.4
SGLang	512	64	shared_prefi	100.0	100.0	129.0	156.6	13.9	15.3
SGLang	512	128	random	100.0	100.0	92.6	118.1	12.8	13.4
SGLang	512	128	shared_prefi	100.0	100.0	129.5	152.5	14.0	15.3
SGLang	1024	1	random	100.0	100.0	49.8	71.4	9.1	10.0
SGLang	1024	1	shared_prefi	100.0	100.0	33.4	35.8	9.4	10.2
SGLang	1024	2	random	100.0	100.0	54.4	56.8	9.6	10.6
SGLang	1024	2	shared_prefi	100.0	100.0	53.7	56.0	9.9	10.8
SGLang	1024	4	random	100.0	100.0	67.9	112.2	9.9	10.9
SGLang	1024	4	shared_prefi	100.0	100.0	67.7	71.2	10.2	11.3
SGLang	1024	8	random	100.0	100.0	76.6	103.0	10.1	10.9
SGLang	1024	8	shared_prefi	100.0	100.0	72.2	100.8	10.6	11.7
SGLang	1024	16	random	100.0	100.0	107.6	137.4	11.1	12.3
SGLang	1024	16	shared_prefi	100.0	100.0	106.0	135.1	11.6	13.0
SGLang	1024	32	random	100.0	100.0	129.7	151.5	13.2	14.5
SGLang	1024	32	shared_prefi	100.0	100.0	130.4	155.6	13.9	15.3
SGLang	1024	64	random	100.0	100.0	128.7	156.8	13.1	14.7
SGLang	1024	64	shared_prefi	100.0	100.0	128.5	150.8	13.9	15.3
SGLang	1024	128	random	100.0	100.0	128.8	151.5	13.1	14.7

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	1024	128	shared_prefi	100.0	100.0	129.3	153.8	13.9	15.3
SGLang	2048	1	random	100.0	100.0	69.8	113.6	9.3	10.2
SGLang	2048	1	shared_prefi	100.0	100.0	33.4	36.5	9.4	10.2
SGLang	2048	2	random	100.0	100.0	57.9	62.9	9.9	11.0
SGLang	2048	2	shared_prefi	100.0	100.0	53.5	54.7	9.9	10.8
SGLang	2048	4	random	100.0	100.0	73.4	79.5	10.1	11.4
SGLang	2048	4	shared_prefi	100.0	100.0	66.8	69.3	10.3	11.3
SGLang	2048	8	random	100.0	100.0	87.7	110.6	10.7	11.6
SGLang	2048	8	shared_prefi	100.0	100.0	94.6	116.3	10.6	11.8
SGLang	2048	16	random	100.0	100.0	124.2	145.5	12.2	13.1
SGLang	2048	16	shared_prefi	100.0	100.0	101.9	129.3	11.6	13.0
SGLang	2048	32	random	100.0	100.0	139.9	172.6	15.2	16.7
SGLang	2048	32	shared_prefi	100.0	100.0	131.2	150.9	13.9	15.3
SGLang	2048	64	random	100.0	100.0	147.1	177.4	15.2	16.6
SGLang	2048	64	shared_prefi	100.0	100.0	129.1	151.9	13.9	15.3
SGLang	2048	128	random	100.0	100.0	147.9	172.7	15.2	16.6
SGLang	2048	128	shared_prefi	100.0	100.0	131.7	151.8	13.9	15.3
SGLang	4096	1	random	100.0	100.0	119.0	189.3	9.5	10.2
SGLang	4096	1	shared_prefi	100.0	100.0	33.4	35.6	9.4	10.2
SGLang	4096	2	random	100.0	100.0	64.8	68.3	10.1	11.0
SGLang	4096	2	shared_prefi	100.0	100.0	53.6	56.1	9.9	10.8
SGLang	4096	4	random	100.0	100.0	94.3	107.8	10.6	11.2
SGLang	4096	4	shared_prefi	100.0	100.0	66.7	70.1	10.3	11.3
SGLang	4096	8	random	100.0	100.0	108.7	132.1	11.6	12.3
SGLang	4096	8	shared_prefi	100.0	100.0	73.2	100.7	10.5	11.7
SGLang	4096	16	random	100.0	100.0	162.9	198.7	13.9	15.0
SGLang	4096	16	shared_prefi	100.0	100.0	102.1	131.5	11.6	13.0
SGLang	4096	32	random	100.0	100.0	189.0	220.9	18.8	20.3
SGLang	4096	32	shared_prefi	100.0	100.0	122.2	146.2	13.9	15.3
SGLang	4096	64	random	100.0	100.0	190.7	213.3	18.9	20.4
SGLang	4096	64	shared_prefi	100.0	100.0	119.6	140.4	13.9	15.3
SGLang	4096	128	random	100.0	100.0	187.9	211.8	18.9	20.4
SGLang	4096	128	shared_prefi	100.0	100.0	130.2	157.4	13.9	15.3
SGLang	8192	1	random	100.0	100.0	219.0	384.1	9.8	11.0
SGLang	8192	1	shared_prefi	100.0	100.0	33.5	36.9	9.4	10.2

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	8192	2	random	100.0	100.0	72.8	84.5	10.7	12.0
SGLang	8192	2	shared_prefi	100.0	100.0	53.4	56.0	9.9	11.0
SGLang	8192	4	random	100.0	100.0	104.8	122.1	11.4	12.7
SGLang	8192	4	shared_prefi	100.0	100.0	66.1	68.8	10.3	11.3
SGLang	8192	8	random	100.0	100.0	132.3	165.8	13.4	15.1
SGLang	8192	8	shared_prefi	100.0	100.0	73.6	100.9	10.6	11.7
SGLang	8192	16	random	100.0	100.0	222.6	252.6	17.6	19.9
SGLang	8192	16	shared_prefi	100.0	100.0	111.4	143.1	11.6	12.9
SGLang	8192	32	random	100.0	100.0	265.9	289.4	26.3	29.1
SGLang	8192	32	shared_prefi	100.0	100.0	128.6	147.5	13.9	15.3
SGLang	8192	64	random	100.0	100.0	266.6	293.9	26.3	29.1
SGLang	8192	64	shared_prefi	100.0	100.0	130.5	155.7	13.8	15.3
SGLang	8192	128	random	100.0	100.0	276.9	295.7	26.3	29.1
SGLang	8192	128	shared_prefi	100.0	100.0	126.1	151.2	13.8	15.3
SGLang	16384	1	random	100.0	0.0	824.9	833.3	10.4	11.2
SGLang	16384	1	shared_prefi	100.0	100.0	33.3	35.7	9.4	10.2
SGLang	16384	2	random	100.0	100.0	90.9	116.1	11.6	12.7
SGLang	16384	2	shared_prefi	100.0	100.0	54.6	56.3	9.9	10.8
SGLang	16384	4	random	100.0	100.0	125.6	176.1	13.3	14.5
SGLang	16384	4	shared_prefi	100.0	100.0	67.3	72.0	10.2	11.3
SGLang	16384	8	random	100.0	100.0	193.2	235.0	17.2	18.9
SGLang	16384	8	shared_prefi	100.0	100.0	73.6	99.5	10.6	11.7
SGLang	16384	16	random	100.0	100.0	343.0	382.7	25.2	27.2
SGLang	16384	16	shared_prefi	100.0	100.0	107.5	138.5	11.6	12.8
SGLang	16384	32	random	100.0	0.0	419.3	483.6	41.2	43.7
SGLang	16384	32	shared_prefi	100.0	100.0	130.6	151.9	13.9	15.3
SGLang	16384	64	random	100.0	0.0	419.9	465.8	41.4	45.2
SGLang	16384	64	shared_prefi	100.0	100.0	120.9	147.0	13.9	15.3
SGLang	16384	128	random	100.0	0.0	422.7	468.8	41.3	44.0
SGLang	16384	128	shared_prefi	100.0	100.0	122.6	144.4	13.9	15.3
SGLang	32768	1	random	100.0	0.0	1917.5	1922.4	11.4	12.2
SGLang	32768	1	shared_prefi	100.0	100.0	33.5	35.9	9.4	10.2
SGLang	32768	2	random	100.0	100.0	128.6	149.8	13.4	14.4
SGLang	32768	2	shared_prefi	100.0	100.0	53.9	59.3	9.9	11.0
SGLang	32768	4	random	100.0	100.0	138.9	235.7	17.3	19.0

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	32768	4	shared_prefi	100.0	100.0	67.5	70.9	10.3	11.3
SGLang	32768	8	random	100.0	100.0	161.9	383.2	24.7	27.2
SGLang	32768	8	shared_prefi	100.0	100.0	94.2	142.4	10.6	11.7
SGLang	32768	16	random	100.0	20.0	637.4	722.6	39.0	42.9
SGLang	32768	16	shared_prefi	100.0	100.0	102.9	134.0	11.5	13.0
SGLang	32768	32	random	100.0	0.0	769.7	857.2	69.8	76.6
SGLang	32768	32	shared_prefi	100.0	100.0	130.9	157.0	13.9	15.3
SGLang	32768	64	random	100.0	0.0	778.0	859.6	69.5	74.7
SGLang	32768	64	shared_prefi	100.0	100.0	130.1	154.2	13.9	15.3
SGLang	32768	128	random	100.0	0.0	794.1	829.1	68.5	72.9
SGLang	32768	128	shared_prefi	100.0	100.0	127.3	153.4	13.9	15.3
vLLM	512	1	random	100.0	100.0	58.3	67.4	9.9	10.5
vLLM	512	1	shared_prefi	100.0	100.0	44.9	96.2	10.1	10.9
vLLM	512	2	random	100.0	100.0	56.0	64.4	9.8	10.7
vLLM	512	2	shared_prefi	100.0	100.0	59.1	73.0	10.0	11.2
vLLM	512	4	random	100.0	100.0	63.2	83.6	10.2	10.9
vLLM	512	4	shared_prefi	100.0	100.0	71.1	83.5	10.2	11.5
vLLM	512	8	random	100.0	100.0	93.3	103.8	10.3	11.0
vLLM	512	8	shared_prefi	100.0	100.0	107.9	121.7	10.7	11.9
vLLM	512	16	random	100.0	100.0	148.1	154.2	10.8	11.5
vLLM	512	16	shared_prefi	100.0	100.0	210.7	216.4	11.3	12.8
vLLM	512	32	random	100.0	100.0	167.0	174.8	12.2	13.0
vLLM	512	32	shared_prefi	100.0	100.0	174.6	184.9	13.6	15.0
vLLM	512	64	random	100.0	100.0	182.5	188.9	12.3	13.1
vLLM	512	64	shared_prefi	100.0	100.0	167.4	177.9	13.5	15.0
vLLM	512	128	random	100.0	100.0	176.0	181.1	12.2	13.0
vLLM	512	128	shared_prefi	100.0	100.0	176.7	187.9	13.4	15.0
vLLM	1024	1	random	100.0	100.0	62.9	80.3	9.6	10.3
vLLM	1024	1	shared_prefi	100.0	100.0	42.5	53.1	10.0	11.1
vLLM	1024	2	random	100.0	100.0	52.0	73.2	9.8	10.7
vLLM	1024	2	shared_prefi	100.0	100.0	58.6	69.1	10.0	11.2
vLLM	1024	4	random	100.0	100.0	69.6	78.8	9.9	11.3
vLLM	1024	4	shared_prefi	100.0	100.0	72.2	82.1	10.2	11.6
vLLM	1024	8	random	100.0	100.0	104.8	119.2	10.2	10.9
vLLM	1024	8	shared_prefi	100.0	100.0	99.3	126.7	10.7	11.9

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	1024	16	random	100.0	100.0	153.9	164.9	11.1	12.1
vLLM	1024	16	shared_prefi	100.0	100.0	125.0	168.7	11.4	12.9
vLLM	1024	32	random	100.0	100.0	187.8	200.9	13.0	13.8
vLLM	1024	32	shared_prefi	100.0	100.0	176.7	185.7	13.5	15.0
vLLM	1024	64	random	100.0	100.0	178.0	185.2	12.9	13.9
vLLM	1024	64	shared_prefi	100.0	100.0	181.9	191.0	13.4	15.0
vLLM	1024	128	random	100.0	100.0	184.3	195.0	12.9	14.2
vLLM	1024	128	shared_prefi	100.0	100.0	185.8	194.9	13.4	15.0
vLLM	2048	1	random	100.0	100.0	82.1	124.1	9.8	10.5
vLLM	2048	1	shared_prefi	100.0	100.0	44.0	54.5	10.0	11.1
vLLM	2048	2	random	100.0	100.0	66.1	75.2	10.1	11.3
vLLM	2048	2	shared_prefi	100.0	100.0	60.4	71.5	10.0	11.2
vLLM	2048	4	random	100.0	100.0	74.2	106.9	10.3	11.5
vLLM	2048	4	shared_prefi	100.0	100.0	69.9	82.6	10.2	11.6
vLLM	2048	8	random	100.0	100.0	108.4	133.6	10.8	11.5
vLLM	2048	8	shared_prefi	100.0	100.0	90.2	122.1	10.7	11.9
vLLM	2048	16	random	100.0	100.0	190.4	202.2	12.0	12.8
vLLM	2048	16	shared_prefi	100.0	100.0	157.3	168.1	11.3	12.8
vLLM	2048	32	random	100.0	100.0	221.5	234.8	15.0	17.9
vLLM	2048	32	shared_prefi	100.0	100.0	176.6	189.8	13.5	15.0
vLLM	2048	64	random	100.0	100.0	218.9	232.5	14.9	17.8
vLLM	2048	64	shared_prefi	100.0	100.0	184.5	191.6	13.4	15.0
vLLM	2048	128	random	100.0	100.0	217.2	226.3	14.9	17.8
vLLM	2048	128	shared_prefi	100.0	100.0	190.8	199.7	13.4	15.0
vLLM	4096	1	random	100.0	100.0	121.1	198.4	10.1	11.1
vLLM	4096	1	shared_prefi	100.0	100.0	44.6	48.4	10.0	11.1
vLLM	4096	2	random	100.0	100.0	75.1	84.2	10.3	11.4
vLLM	4096	2	shared_prefi	100.0	100.0	59.1	71.6	10.0	11.2
vLLM	4096	4	random	100.0	100.0	86.4	120.6	10.7	11.7
vLLM	4096	4	shared_prefi	100.0	100.0	69.9	80.6	10.2	11.6
vLLM	4096	8	random	100.0	100.0	140.2	174.8	11.6	12.5
vLLM	4096	8	shared_prefi	100.0	100.0	106.1	124.6	10.7	11.9
vLLM	4096	16	random	100.0	100.0	229.2	245.5	13.9	14.8
vLLM	4096	16	shared_prefi	100.0	100.0	163.7	172.7	11.3	12.8
vLLM	4096	32	random	100.0	100.0	264.1	282.7	18.6	19.4

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	4096	32	shared_prefi	100.0	100.0	174.9	181.5	13.4	15.0
vLLM	4096	64	random	100.0	100.0	236.1	322.2	18.7	19.5
vLLM	4096	64	shared_prefi	100.0	100.0	187.5	196.1	13.4	15.0
vLLM	4096	128	random	100.0	100.0	234.2	310.3	18.7	19.6
vLLM	4096	128	shared_prefi	100.0	100.0	178.7	187.9	13.4	15.0
vLLM	8192	1	random	100.0	100.0	229.1	386.4	10.3	11.7
vLLM	8192	1	shared_prefi	100.0	100.0	43.1	50.8	10.0	11.1
vLLM	8192	2	random	100.0	100.0	86.5	108.5	10.8	12.1
vLLM	8192	2	shared_prefi	100.0	100.0	59.0	69.0	10.0	11.2
vLLM	8192	4	random	100.0	100.0	109.1	126.2	11.6	13.2
vLLM	8192	4	shared_prefi	100.0	100.0	70.3	79.3	10.2	11.5
vLLM	8192	8	random	100.0	100.0	112.6	211.3	13.6	15.2
vLLM	8192	8	shared_prefi	100.0	100.0	106.0	119.1	10.5	11.9
vLLM	8192	16	random	100.0	100.0	255.0	378.6	17.3	19.1
vLLM	8192	16	shared_prefi	100.0	100.0	152.8	160.4	11.3	12.8
vLLM	8192	32	random	100.0	100.0	283.1	425.0	25.9	29.7
vLLM	8192	32	shared_prefi	100.0	100.0	165.1	220.5	13.4	15.0
vLLM	8192	64	random	100.0	100.0	282.8	416.0	26.0	29.7
vLLM	8192	64	shared_prefi	100.0	100.0	172.8	183.2	13.4	15.0
vLLM	8192	128	random	100.0	100.0	271.5	400.4	25.8	29.6
vLLM	8192	128	shared_prefi	100.0	100.0	173.7	180.5	13.5	15.0
vLLM	16384	1	random	100.0	0.0	810.0	819.8	10.6	11.6
vLLM	16384	1	shared_prefi	100.0	100.0	42.8	48.8	10.0	11.1
vLLM	16384	2	random	100.0	100.0	112.5	146.8	11.8	13.5
vLLM	16384	2	shared_prefi	100.0	100.0	59.1	67.1	10.0	11.2
vLLM	16384	4	random	100.0	100.0	144.0	213.0	13.4	15.5
vLLM	16384	4	shared_prefi	100.0	100.0	70.6	80.6	10.3	11.5
vLLM	16384	8	random	100.0	100.0	163.0	284.8	16.9	18.8
vLLM	16384	8	shared_prefi	100.0	100.0	105.2	122.9	10.5	11.9
vLLM	16384	16	random	100.0	95.0	283.8	502.1	24.1	27.4
vLLM	16384	16	shared_prefi	100.0	100.0	151.1	158.8	11.3	12.8
vLLM	16384	32	random	100.0	0.0	456.9	661.7	38.9	43.8
vLLM	16384	32	shared_prefi	100.0	100.0	181.5	187.8	13.4	15.0
vLLM	16384	64	random	100.0	0.0	515.6	572.7	39.6	43.9
vLLM	16384	64	shared_prefi	100.0	100.0	173.9	182.5	13.4	15.0

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	16384	128	random	100.0	0.0	487.6	573.8	39.6	43.9
vLLM	16384	128	shared_prefi	100.0	100.0	182.2	189.6	13.5	15.0
vLLM	32768	1	random	100.0	0.0	1903.5	1919.8	11.9	13.0
vLLM	32768	1	shared_prefi	100.0	100.0	43.6	48.6	10.0	11.1
vLLM	32768	2	random	100.0	100.0	163.1	207.5	13.4	15.1
vLLM	32768	2	shared_prefi	100.0	100.0	58.8	68.5	10.0	11.2
vLLM	32768	4	random	100.0	100.0	166.0	271.0	17.1	20.0
vLLM	32768	4	shared_prefi	100.0	100.0	70.1	84.0	10.2	11.5
vLLM	32768	8	random	100.0	85.0	247.2	510.5	24.2	26.7
vLLM	32768	8	shared_prefi	100.0	100.0	104.2	109.4	10.6	11.9
vLLM	32768	16	random	100.0	20.0	517.2	901.4	38.8	42.8
vLLM	32768	16	shared_prefi	100.0	100.0	165.7	175.5	11.3	12.8
vLLM	32768	32	random	100.0	0.0	745.3	1139.9	68.8	74.4
vLLM	32768	32	shared_prefi	100.0	100.0	188.1	194.0	13.4	15.0
vLLM	32768	64	random	100.0	0.0	710.2	1108.8	69.1	74.6
vLLM	32768	64	shared_prefi	100.0	100.0	183.6	192.9	13.5	15.0
vLLM	32768	128	random	100.0	0.0	668.5	1114.4	69.1	77.7
vLLM	32768	128	shared_prefi	100.0	100.0	180.8	185.6	13.5	15.0

Phase A — SGLang Default (112 cells)

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	512	1	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	1	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	2	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	2	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	4	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	4	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	8	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	8	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	16	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	16	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	32	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	32	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	64	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	64	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	512	128	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	512	128	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	1	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	1	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	2	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	2	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	4	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	4	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	8	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	8	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	16	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	16	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	32	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	32	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	64	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	64	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	128	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	1024	128	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	1	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	1	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	2	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	2	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	4	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	4	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	8	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	8	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	16	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	16	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	32	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	32	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	64	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	64	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	128	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	2048	128	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	1	random	0.0	0.0	0.0	0.0	0.0	0.0

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	4096	1	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	2	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	2	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	4	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	4	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	8	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	8	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	16	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	16	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	32	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	32	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	64	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	64	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	128	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	4096	128	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	1	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	1	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	2	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	2	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	4	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	4	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	8	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	8	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	16	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	16	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	32	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	32	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	64	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	64	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	128	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	8192	128	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	1	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	1	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	2	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	2	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	16384	4	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	4	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	8	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	8	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	16	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	16	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	32	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	32	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	64	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	64	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	128	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	16384	128	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	1	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	1	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	2	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	2	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	4	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	4	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	8	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	8	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	16	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	16	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	32	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	32	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	64	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	64	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	128	random	0.0	0.0	0.0	0.0	0.0	0.0
SGLang	32768	128	shared_prefi	0.0	0.0	0.0	0.0	0.0	0.0

Phase B — vLLM Default (112 cells)

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	512	1	random	100.0	100.0	41.0	48.8	9.8	10.7
vLLM	512	1	shared_prefi	100.0	100.0	42.3	88.4	10.1	11.0
vLLM	512	2	random	100.0	100.0	55.7	65.1	9.9	10.3
vLLM	512	2	shared_prefi	100.0	100.0	57.3	69.8	10.1	11.2

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	512	4	random	100.0	100.0	66.0	74.8	10.1	10.7
vLLM	512	4	shared_prefi	100.0	100.0	69.5	80.8	10.2	11.5
vLLM	512	8	random	100.0	100.0	91.1	97.6	10.3	11.1
vLLM	512	8	shared_prefi	100.0	100.0	97.9	113.8	10.7	11.9
vLLM	512	16	random	100.0	100.0	122.0	128.5	10.7	11.6
vLLM	512	16	shared_prefi	100.0	100.0	150.6	164.6	11.2	12.8
vLLM	512	32	random	100.0	100.0	121.4	131.0	12.3	13.6
vLLM	512	32	shared_prefi	100.0	100.0	179.1	187.6	13.4	15.0
vLLM	512	64	random	100.0	100.0	148.1	156.7	12.4	13.6
vLLM	512	64	shared_prefi	100.0	100.0	154.2	163.6	13.4	15.0
vLLM	512	128	random	100.0	100.0	150.8	158.4	12.3	13.5
vLLM	512	128	shared_prefi	100.0	100.0	174.8	187.2	13.5	15.0
vLLM	1024	1	random	100.0	100.0	42.8	46.6	9.6	10.8
vLLM	1024	1	shared_prefi	100.0	100.0	41.5	43.1	10.0	11.2
vLLM	1024	2	random	100.0	100.0	59.3	67.9	9.8	11.0
vLLM	1024	2	shared_prefi	100.0	100.0	58.7	65.2	10.1	11.2
vLLM	1024	4	random	100.0	100.0	70.0	96.7	9.9	11.2
vLLM	1024	4	shared_prefi	100.0	100.0	70.7	82.8	10.2	11.5
vLLM	1024	8	random	100.0	100.0	76.9	104.2	10.3	11.1
vLLM	1024	8	shared_prefi	100.0	100.0	104.6	116.4	10.5	11.9
vLLM	1024	16	random	100.0	100.0	144.7	151.4	11.1	12.2
vLLM	1024	16	shared_prefi	100.0	100.0	151.7	161.3	11.3	12.8
vLLM	1024	32	random	100.0	100.0	162.1	171.1	13.0	14.3
vLLM	1024	32	shared_prefi	100.0	100.0	171.3	177.2	13.4	15.0
vLLM	1024	64	random	100.0	100.0	157.7	166.2	12.9	14.3
vLLM	1024	64	shared_prefi	100.0	100.0	176.7	185.5	13.4	15.0
vLLM	1024	128	random	100.0	100.0	178.3	191.8	13.0	14.3
vLLM	1024	128	shared_prefi	100.0	100.0	182.3	188.1	13.4	15.0
vLLM	2048	1	random	100.0	100.0	47.6	55.9	9.8	11.5
vLLM	2048	1	shared_prefi	100.0	100.0	41.8	45.4	10.0	11.2
vLLM	2048	2	random	100.0	100.0	63.7	73.9	10.0	11.7
vLLM	2048	2	shared_prefi	100.0	100.0	58.7	67.2	10.1	11.2
vLLM	2048	4	random	100.0	100.0	76.5	95.0	10.4	11.1
vLLM	2048	4	shared_prefi	100.0	100.0	69.9	78.7	10.3	11.6
vLLM	2048	8	random	100.0	100.0	99.2	124.7	10.9	11.8

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	2048	8	shared_prefi	100.0	100.0	105.3	117.6	10.7	11.9
vLLM	2048	16	random	100.0	100.0	179.3	196.3	12.0	13.2
vLLM	2048	16	shared_prefi	100.0	100.0	152.4	192.7	11.2	12.8
vLLM	2048	32	random	100.0	100.0	205.9	214.6	14.9	16.8
vLLM	2048	32	shared_prefi	100.0	100.0	171.5	186.2	13.4	15.0
vLLM	2048	64	random	100.0	100.0	195.6	205.8	14.9	16.6
vLLM	2048	64	shared_prefi	100.0	100.0	176.9	186.9	13.3	15.0
vLLM	2048	128	random	100.0	100.0	169.0	181.1	14.8	16.6
vLLM	2048	128	shared_prefi	100.0	100.0	178.7	188.2	13.5	15.0
vLLM	4096	1	random	100.0	100.0	57.3	197.1	10.1	11.3
vLLM	4096	1	shared_prefi	100.0	100.0	41.5	44.7	10.0	11.2
vLLM	4096	2	random	100.0	100.0	75.4	85.0	10.2	10.8
vLLM	4096	2	shared_prefi	100.0	100.0	57.6	67.7	10.1	11.2
vLLM	4096	4	random	100.0	100.0	88.1	122.3	10.7	11.3
vLLM	4096	4	shared_prefi	100.0	100.0	69.2	82.5	10.2	11.6
vLLM	4096	8	random	100.0	100.0	106.6	173.2	11.6	12.3
vLLM	4096	8	shared_prefi	100.0	100.0	99.5	118.8	10.7	11.9
vLLM	4096	16	random	100.0	100.0	236.1	256.0	13.7	15.0
vLLM	4096	16	shared_prefi	100.0	100.0	154.6	172.5	11.2	12.8
vLLM	4096	32	random	100.0	100.0	251.3	280.9	18.5	21.0
vLLM	4096	32	shared_prefi	100.0	100.0	178.6	184.6	13.4	15.0
vLLM	4096	64	random	100.0	100.0	246.2	304.3	18.5	20.3
vLLM	4096	64	shared_prefi	100.0	100.0	175.8	191.7	13.4	15.0
vLLM	4096	128	random	100.0	100.0	248.8	297.1	18.5	21.0
vLLM	4096	128	shared_prefi	100.0	100.0	178.7	187.2	13.3	15.0
vLLM	8192	1	random	100.0	100.0	384.2	389.7	10.3	11.9
vLLM	8192	1	shared_prefi	100.0	100.0	41.4	43.8	10.0	11.2
vLLM	8192	2	random	100.0	100.0	81.3	108.2	10.8	11.7
vLLM	8192	2	shared_prefi	100.0	100.0	58.5	70.1	10.1	11.2
vLLM	8192	4	random	100.0	100.0	107.7	151.7	11.6	12.5
vLLM	8192	4	shared_prefi	100.0	100.0	65.5	80.5	10.3	11.6
vLLM	8192	8	random	100.0	100.0	125.0	228.0	13.6	14.7
vLLM	8192	8	shared_prefi	100.0	100.0	98.2	115.7	10.7	11.9
vLLM	8192	16	random	100.0	100.0	277.7	369.3	17.5	19.7
vLLM	8192	16	shared_prefi	100.0	100.0	151.5	165.9	11.3	12.8

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	8192	32	random	100.0	100.0	327.2	432.4	25.9	28.8
vLLM	8192	32	shared_prefi	100.0	100.0	175.2	180.3	13.4	15.0
vLLM	8192	64	random	100.0	100.0	300.0	426.8	25.7	28.5
vLLM	8192	64	shared_prefi	100.0	100.0	178.9	184.2	13.4	15.0
vLLM	8192	128	random	100.0	100.0	310.3	400.9	25.7	28.5
vLLM	8192	128	shared_prefi	100.0	100.0	179.5	187.9	13.4	15.0
vLLM	16384	1	random	100.0	0.0	808.6	818.0	10.7	11.7
vLLM	16384	1	shared_prefi	100.0	100.0	41.2	43.5	10.0	11.2
vLLM	16384	2	random	100.0	100.0	102.9	146.6	11.8	12.8
vLLM	16384	2	shared_prefi	100.0	100.0	57.0	64.8	10.1	11.2
vLLM	16384	4	random	100.0	100.0	134.7	180.4	13.5	14.7
vLLM	16384	4	shared_prefi	100.0	100.0	70.0	79.7	10.3	11.6
vLLM	16384	8	random	100.0	100.0	173.1	318.5	16.6	18.6
vLLM	16384	8	shared_prefi	100.0	100.0	104.8	111.3	10.7	11.9
vLLM	16384	16	random	100.0	95.0	307.7	494.7	24.4	27.3
vLLM	16384	16	shared_prefi	100.0	100.0	153.2	165.8	11.4	12.8
vLLM	16384	32	random	100.0	0.0	381.6	616.2	39.7	43.9
vLLM	16384	32	shared_prefi	100.0	100.0	179.8	187.6	13.5	15.0
vLLM	16384	64	random	100.0	0.0	433.0	538.8	39.7	43.7
vLLM	16384	64	shared_prefi	100.0	100.0	175.6	188.1	13.3	15.0
vLLM	16384	128	random	100.0	0.0	398.0	652.0	39.6	42.4
vLLM	16384	128	shared_prefi	100.0	100.0	179.1	187.4	13.4	15.0
vLLM	32768	1	random	100.0	0.0	1911.6	1921.3	11.9	13.0
vLLM	32768	1	shared_prefi	100.0	100.0	41.2	42.9	10.0	11.2
vLLM	32768	2	random	100.0	100.0	148.6	175.8	13.3	15.2
vLLM	32768	2	shared_prefi	100.0	100.0	58.2	66.0	10.1	11.2
vLLM	32768	4	random	100.0	100.0	165.9	286.9	17.2	18.8
vLLM	32768	4	shared_prefi	100.0	100.0	70.6	79.8	10.2	11.6
vLLM	32768	8	random	100.0	100.0	208.3	483.6	24.7	26.1
vLLM	32768	8	shared_prefi	100.0	100.0	98.4	119.4	10.7	11.9
vLLM	32768	16	random	100.0	20.0	489.1	864.8	39.1	43.1
vLLM	32768	16	shared_prefi	100.0	100.0	181.6	190.5	11.3	12.8
vLLM	32768	32	random	100.0	0.0	729.0	1072.0	69.2	76.7
vLLM	32768	32	shared_prefi	100.0	100.0	179.3	193.2	13.4	15.0
vLLM	32768	64	random	100.0	0.0	728.2	1145.6	69.1	79.8

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	32768	64	shared_prefi	100.0	100.0	180.2	186.7	13.4	15.0
vLLM	32768	128	random	100.0	0.0	726.7	1094.7	69.4	79.6
vLLM	32768	128	shared_prefi	100.0	100.0	177.0	182.3	13.4	15.0

Phase D — SGLang Long-Context

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	65536	1	random	100.0	0.0	20913.9	21057.8	39.7	45.3
SGLang	65536	1	shared_prefi	100.0	100.0	41.8	100.1	9.6	10.6
SGLang	65536	2	random	100.0	0.0	817.1	1117.5	42.1	47.4
SGLang	65536	2	shared_prefi	100.0	100.0	63.6	65.3	10.1	11.0
SGLang	65536	4	random	100.0	0.0	1013.1	1883.7	47.4	51.6
SGLang	65536	4	shared_prefi	100.0	100.0	69.1	86.2	10.1	11.0
SGLang	65536	8	random	100.0	0.0	1607.6	2035.9	55.9	63.7
SGLang	65536	8	shared_prefi	100.0	100.0	82.7	103.6	10.6	11.2
SGLang	131072	1	random	100.0	0.0	55559.5	55838.8	71.4	84.1
SGLang	131072	1	shared_prefi	100.0	100.0	38.3	44.3	9.6	10.3
SGLang	131072	2	random	100.0	0.0	1585.1	2158.9	70.6	79.6
SGLang	131072	2	shared_prefi	100.0	100.0	59.2	61.4	9.9	11.0
SGLang	131072	4	random	100.0	0.0	1994.9	2216.6	80.0	90.8
SGLang	131072	4	shared_prefi	100.0	100.0	67.6	69.0	10.1	11.0
SGLang	131072	8	random	100.0	0.0	3139.1	4113.0	102.6	115.3
SGLang	131072	8	shared_prefi	100.0	100.0	77.6	117.3	10.6	11.1

Phase F — SGLang TP=2 Default

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	512	1	random	100.0	100.0	94.2	101.2	3.4	3.9
SGLang	512	1	shared_prefi	100.0	100.0	59.2	69.0	3.6	4.0
SGLang	512	2	random	100.0	100.0	79.7	84.8	3.5	4.1
SGLang	512	2	shared_prefi	100.0	100.0	100.6	107.6	3.9	4.3
SGLang	512	4	random	100.0	100.0	87.5	91.6	3.9	4.4
SGLang	512	4	shared_prefi	100.0	100.0	86.1	89.5	4.2	4.6
SGLang	512	8	random	100.0	100.0	111.3	118.4	4.5	4.8
SGLang	512	8	shared_prefi	100.0	100.0	94.6	98.3	4.8	5.2
SGLang	512	16	random	100.0	100.0	140.5	149.0	5.8	6.2
SGLang	512	16	shared_prefi	100.0	100.0	126.3	134.9	5.6	6.3

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	512	32	random	100.0	100.0	162.2	168.9	6.2	6.7
SGLang	512	32	shared_prefi	100.0	100.0	118.5	147.5	6.6	7.2
SGLang	512	64	random	100.0	100.0	148.3	161.0	6.3	6.7
SGLang	512	64	shared_prefi	100.0	100.0	142.3	157.8	6.6	7.3
SGLang	512	128	random	100.0	100.0	147.8	155.8	6.2	6.7
SGLang	512	128	shared_prefi	100.0	100.0	137.6	154.1	6.2	6.8
SGLang	1024	1	random	100.0	100.0	55.0	60.4	3.4	3.7
SGLang	1024	1	shared_prefi	100.0	100.0	60.4	67.2	3.6	4.0
SGLang	1024	2	random	100.0	100.0	82.0	87.1	3.5	3.9
SGLang	1024	2	shared_prefi	100.0	100.0	99.2	102.3	3.9	4.3
SGLang	1024	4	random	100.0	100.0	93.2	135.8	3.8	4.1
SGLang	1024	4	shared_prefi	100.0	100.0	86.1	89.6	4.2	4.9
SGLang	1024	8	random	100.0	100.0	117.9	125.6	4.4	4.8
SGLang	1024	8	shared_prefi	100.0	100.0	98.2	110.4	4.7	5.7
SGLang	1024	16	random	100.0	100.0	139.5	151.4	5.8	6.5
SGLang	1024	16	shared_prefi	100.0	100.0	125.5	133.2	5.5	6.2
SGLang	1024	32	random	100.0	100.0	171.1	183.0	6.3	7.1
SGLang	1024	32	shared_prefi	100.0	100.0	130.5	144.4	6.6	7.2
SGLang	1024	64	random	100.0	100.0	185.7	238.6	6.6	7.4
SGLang	1024	64	shared_prefi	100.0	100.0	135.1	150.7	6.7	7.3
SGLang	1024	128	random	100.0	100.0	165.8	184.4	6.6	7.3
SGLang	1024	128	shared_prefi	100.0	100.0	129.9	142.8	5.8	6.4
SGLang	2048	1	random	100.0	100.0	100.2	102.8	3.4	3.8
SGLang	2048	1	shared_prefi	100.0	100.0	59.0	76.0	3.6	4.0
SGLang	2048	2	random	100.0	100.0	83.9	88.7	3.6	4.0
SGLang	2048	2	shared_prefi	100.0	100.0	98.4	102.5	3.9	4.3
SGLang	2048	4	random	100.0	100.0	94.8	97.6	4.0	4.4
SGLang	2048	4	shared_prefi	100.0	100.0	86.2	89.7	4.2	4.8
SGLang	2048	8	random	100.0	100.0	118.5	177.6	4.7	5.3
SGLang	2048	8	shared_prefi	100.0	100.0	92.4	106.5	4.7	5.8
SGLang	2048	16	random	100.0	100.0	213.9	221.7	6.2	6.8
SGLang	2048	16	shared_prefi	100.0	100.0	162.6	169.8	5.7	6.3
SGLang	2048	32	random	100.0	100.0	192.5	228.7	7.3	8.0
SGLang	2048	32	shared_prefi	100.0	100.0	124.7	144.4	6.5	7.2
SGLang	2048	64	random	100.0	100.0	191.2	218.0	7.3	8.1

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	2048	64	shared_prefi	100.0	100.0	121.5	143.2	6.5	7.1
SGLang	2048	128	random	100.0	100.0	263.2	270.6	7.1	7.9
SGLang	2048	128	shared_prefi	100.0	100.0	132.9	150.4	5.6	6.1
SGLang	4096	1	random	100.0	100.0	103.6	114.6	3.5	3.7
SGLang	4096	1	shared_prefi	100.0	100.0	59.1	63.7	3.6	4.0
SGLang	4096	2	random	100.0	100.0	90.0	93.9	3.6	4.0
SGLang	4096	2	shared_prefi	100.0	100.0	99.6	104.7	3.9	4.3
SGLang	4096	4	random	100.0	100.0	112.0	151.4	4.1	4.7
SGLang	4096	4	shared_prefi	100.0	100.0	85.5	88.8	4.2	5.0
SGLang	4096	8	random	100.0	100.0	168.1	226.6	5.0	5.3
SGLang	4096	8	shared_prefi	100.0	100.0	96.3	106.3	4.7	5.9
SGLang	4096	16	random	100.0	100.0	242.5	325.2	6.9	7.4
SGLang	4096	16	shared_prefi	100.0	100.0	121.8	131.5	5.3	5.9
SGLang	4096	32	random	100.0	100.0	405.4	411.3	8.0	8.6
SGLang	4096	32	shared_prefi	100.0	100.0	131.1	143.7	6.5	7.1
SGLang	4096	64	random	100.0	100.0	408.8	456.8	8.0	8.7
SGLang	4096	64	shared_prefi	100.0	100.0	128.7	149.9	6.5	7.2
SGLang	4096	128	random	100.0	100.0	339.9	431.3	8.3	9.2
SGLang	4096	128	shared_prefi	100.0	100.0	135.3	148.2	5.6	6.1
SGLang	8192	1	random	100.0	100.0	115.7	119.4	3.6	4.0
SGLang	8192	1	shared_prefi	100.0	100.0	59.4	72.1	3.6	4.0
SGLang	8192	2	random	100.0	100.0	132.9	142.3	3.8	4.2
SGLang	8192	2	shared_prefi	100.0	100.0	100.2	106.3	3.9	4.3
SGLang	8192	4	random	100.0	100.0	157.0	233.8	4.4	5.0
SGLang	8192	4	shared_prefi	100.0	100.0	85.6	96.2	4.2	5.0
SGLang	8192	8	random	100.0	100.0	296.1	302.9	5.5	6.5
SGLang	8192	8	shared_prefi	100.0	100.0	97.9	103.7	4.7	5.7
SGLang	8192	16	random	100.0	20.0	583.5	592.9	7.4	8.3
SGLang	8192	16	shared_prefi	100.0	100.0	121.8	130.2	6.2	6.9
SGLang	8192	32	random	100.0	0.0	696.5	711.4	9.8	10.8
SGLang	8192	32	shared_prefi	100.0	100.0	124.6	141.2	6.6	7.2
SGLang	8192	64	random	100.0	0.0	703.8	712.1	10.0	10.9
SGLang	8192	64	shared_prefi	100.0	100.0	121.6	143.0	6.4	7.0
SGLang	8192	128	random	100.0	0.0	711.7	735.7	9.7	10.8
SGLang	8192	128	shared_prefi	100.0	100.0	126.1	157.1	5.7	6.2

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
SGLang	16384	1	random	100.0	100.0	307.1	320.3	3.6	3.9
SGLang	16384	1	shared_prefi	100.0	100.0	59.5	69.3	3.6	4.0
SGLang	16384	2	random	100.0	100.0	173.9	218.3	4.1	4.4
SGLang	16384	2	shared_prefi	100.0	100.0	98.3	102.1	3.9	4.4
SGLang	16384	4	random	100.0	100.0	276.7	327.8	4.8	5.7
SGLang	16384	4	shared_prefi	100.0	100.0	84.3	91.3	4.2	4.8
SGLang	16384	8	random	100.0	20.0	527.8	586.8	6.5	7.4
SGLang	16384	8	shared_prefi	100.0	100.0	94.8	99.9	4.7	5.5
SGLang	16384	16	random	100.0	20.0	1056.4	1132.3	9.8	11.4
SGLang	16384	16	shared_prefi	100.0	100.0	123.3	130.2	5.6	6.3
SGLang	16384	32	random	100.0	0.0	1363.0	1370.9	12.8	13.7
SGLang	16384	32	shared_prefi	100.0	100.0	136.8	150.4	6.6	7.2
SGLang	16384	64	random	100.0	0.0	1343.3	1388.4	12.9	13.7
SGLang	16384	64	shared_prefi	100.0	100.0	124.7	148.1	6.5	7.2
SGLang	16384	128	random	100.0	0.0	1341.4	1391.2	12.9	13.8
SGLang	16384	128	shared_prefi	100.0	100.0	136.2	144.0	5.6	6.2
SGLang	32768	1	random	100.0	0.0	629.6	653.9	4.0	4.4
SGLang	32768	1	shared_prefi	100.0	100.0	59.1	63.5	3.6	4.0
SGLang	32768	2	random	100.0	100.0	223.0	276.1	4.5	5.0
SGLang	32768	2	shared_prefi	100.0	100.0	98.3	106.5	3.9	4.3
SGLang	32768	4	random	100.0	80.0	215.0	563.9	6.2	6.7
SGLang	32768	4	shared_prefi	100.0	100.0	84.3	90.4	4.2	5.0
SGLang	32768	8	random	100.0	60.0	279.5	611.0	9.7	10.6
SGLang	32768	8	shared_prefi	100.0	100.0	98.1	102.9	4.7	5.8
SGLang	32768	16	random	100.0	20.0	1057.7	1094.7	13.8	15.2
SGLang	32768	16	shared_prefi	100.0	100.0	177.1	182.3	6.1	6.9
SGLang	32768	32	random	100.0	0.0	1381.9	1414.4	17.2	21.3
SGLang	32768	32	shared_prefi	100.0	100.0	130.2	146.0	6.6	7.3
SGLang	32768	64	random	100.0	0.0	685.0	732.6	19.6	21.1
SGLang	32768	64	shared_prefi	100.0	100.0	137.5	147.4	6.3	6.9
SGLang	32768	128	random	100.0	0.0	1338.2	1362.2	17.6	18.9
SGLang	32768	128	shared_prefi	100.0	100.0	130.8	142.3	5.5	6.0

Phase F — vLLM TP=2 Default

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	512	1	random	100.0	100.0	26.9	39.8	3.3	3.5
vLLM	512	1	shared_prefi	100.0	100.0	27.2	31.5	3.4	3.7
vLLM	512	2	random	100.0	100.0	34.9	41.1	3.3	3.5
vLLM	512	2	shared_prefi	100.0	100.0	31.6	40.2	3.4	3.8
vLLM	512	4	random	100.0	100.0	45.0	56.1	3.5	3.7
vLLM	512	4	shared_prefi	100.0	100.0	45.4	54.3	3.6	4.0
vLLM	512	8	random	100.0	100.0	60.5	67.8	3.7	4.1
vLLM	512	8	shared_prefi	100.0	100.0	56.0	84.0	3.9	4.3
vLLM	512	16	random	100.0	100.0	89.9	108.0	4.1	4.5
vLLM	512	16	shared_prefi	100.0	100.0	99.9	123.4	4.3	4.8
vLLM	512	32	random	100.0	100.0	98.2	117.8	4.2	4.5
vLLM	512	32	shared_prefi	100.0	100.0	119.5	153.0	4.7	5.1
vLLM	512	64	random	100.0	100.0	110.8	127.3	4.5	5.0
vLLM	512	64	shared_prefi	100.0	100.0	118.9	148.4	4.6	5.1
vLLM	512	128	random	100.0	100.0	113.9	131.8	4.5	4.7
vLLM	512	128	shared_prefi	100.0	100.0	119.3	147.7	4.6	5.1
vLLM	1024	1	random	100.0	100.0	31.6	35.2	3.2	3.9
vLLM	1024	1	shared_prefi	100.0	100.0	31.3	37.0	3.3	3.7
vLLM	1024	2	random	100.0	100.0	36.8	51.1	3.3	3.6
vLLM	1024	2	shared_prefi	100.0	100.0	30.5	39.9	3.4	3.8
vLLM	1024	4	random	100.0	100.0	46.4	72.9	3.4	3.9
vLLM	1024	4	shared_prefi	100.0	100.0	44.3	55.4	3.6	4.0
vLLM	1024	8	random	100.0	100.0	56.4	72.9	3.7	3.9
vLLM	1024	8	shared_prefi	100.0	100.0	51.5	79.0	3.8	4.3
vLLM	1024	16	random	100.0	100.0	97.8	146.1	4.3	4.7
vLLM	1024	16	shared_prefi	100.0	100.0	99.3	127.7	4.3	4.8
vLLM	1024	32	random	100.0	100.0	142.2	159.7	4.7	5.2
vLLM	1024	32	shared_prefi	100.0	100.0	116.3	154.8	4.7	5.2
vLLM	1024	64	random	100.0	100.0	125.9	169.4	4.8	5.3
vLLM	1024	64	shared_prefi	100.0	100.0	125.7	151.1	4.7	5.2
vLLM	1024	128	random	100.0	100.0	121.0	161.8	4.7	5.3
vLLM	1024	128	shared_prefi	100.0	100.0	107.4	142.6	4.7	5.1
vLLM	2048	1	random	100.0	100.0	36.6	43.1	3.3	3.5
vLLM	2048	1	shared_prefi	100.0	100.0	30.7	31.6	3.3	3.7
vLLM	2048	2	random	100.0	100.0	36.7	55.8	3.3	3.6

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	2048	2	shared_prefi	100.0	100.0	33.4	44.2	3.4	3.8
vLLM	2048	4	random	100.0	100.0	46.6	89.1	3.6	3.9
vLLM	2048	4	shared_prefi	100.0	100.0	38.1	55.8	3.6	4.0
vLLM	2048	8	random	100.0	100.0	68.4	107.5	3.9	4.4
vLLM	2048	8	shared_prefi	100.0	100.0	57.3	78.0	3.8	4.3
vLLM	2048	16	random	100.0	100.0	110.3	159.9	4.5	4.9
vLLM	2048	16	shared_prefi	100.0	100.0	104.7	130.5	4.3	4.8
vLLM	2048	32	random	100.0	100.0	136.3	178.4	5.0	5.5
vLLM	2048	32	shared_prefi	100.0	100.0	146.0	177.0	4.7	5.2
vLLM	2048	64	random	100.0	100.0	126.9	168.6	5.0	5.4
vLLM	2048	64	shared_prefi	100.0	100.0	112.2	149.5	4.7	5.2
vLLM	2048	128	random	100.0	100.0	138.6	179.1	5.0	5.5
vLLM	2048	128	shared_prefi	100.0	100.0	114.2	149.6	4.7	5.2
vLLM	4096	1	random	100.0	100.0	43.3	46.7	3.3	3.6
vLLM	4096	1	shared_prefi	100.0	100.0	31.7	32.6	3.3	3.7
vLLM	4096	2	random	100.0	100.0	42.2	53.4	3.4	3.7
vLLM	4096	2	shared_prefi	100.0	100.0	36.4	48.7	3.4	3.8
vLLM	4096	4	random	100.0	100.0	49.0	70.7	3.7	3.9
vLLM	4096	4	shared_prefi	100.0	100.0	39.2	54.0	3.6	4.0
vLLM	4096	8	random	100.0	100.0	71.5	130.0	4.1	4.5
vLLM	4096	8	shared_prefi	100.0	100.0	60.3	83.4	3.9	4.4
vLLM	4096	16	random	100.0	100.0	143.2	221.9	4.9	5.3
vLLM	4096	16	shared_prefi	100.0	100.0	118.2	146.1	5.0	5.7
vLLM	4096	32	random	100.0	100.0	182.7	268.1	5.7	6.1
vLLM	4096	32	shared_prefi	100.0	100.0	108.9	133.5	4.8	5.3
vLLM	4096	64	random	100.0	100.0	170.3	261.0	5.7	6.1
vLLM	4096	64	shared_prefi	100.0	100.0	112.9	136.8	4.8	5.3
vLLM	4096	128	random	100.0	100.0	165.8	250.2	5.7	6.0
vLLM	4096	128	shared_prefi	100.0	100.0	109.3	135.4	4.8	5.4
vLLM	8192	1	random	100.0	100.0	47.3	57.1	3.4	3.9
vLLM	8192	1	shared_prefi	100.0	100.0	31.3	38.5	3.3	3.7
vLLM	8192	2	random	100.0	100.0	56.6	72.0	3.6	3.9
vLLM	8192	2	shared_prefi	100.0	100.0	36.6	44.7	3.4	3.8
vLLM	8192	4	random	100.0	100.0	62.2	103.2	3.9	4.3
vLLM	8192	4	shared_prefi	100.0	100.0	46.2	58.9	3.6	4.0

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	8192	8	random	100.0	100.0	75.9	177.7	4.6	5.1
vLLM	8192	8	shared_prefi	100.0	100.0	71.1	79.8	3.8	4.3
vLLM	8192	16	random	100.0	100.0	197.9	373.3	6.1	6.9
vLLM	8192	16	shared_prefi	100.0	100.0	83.7	116.3	4.3	4.8
vLLM	8192	32	random	100.0	85.0	360.8	522.0	7.4	7.9
vLLM	8192	32	shared_prefi	100.0	100.0	109.7	135.6	4.9	5.4
vLLM	8192	64	random	100.0	100.0	331.5	494.6	7.4	8.1
vLLM	8192	64	shared_prefi	100.0	100.0	105.2	129.7	4.9	5.3
vLLM	8192	128	random	100.0	90.0	378.8	532.6	7.5	8.0
vLLM	8192	128	shared_prefi	100.0	100.0	107.0	131.3	4.8	5.4
vLLM	16384	1	random	100.0	100.0	280.4	289.1	3.5	3.7
vLLM	16384	1	shared_prefi	100.0	100.0	31.6	32.7	3.3	3.7
vLLM	16384	2	random	100.0	100.0	84.0	108.3	3.8	4.1
vLLM	16384	2	shared_prefi	100.0	100.0	35.5	42.6	3.4	3.8
vLLM	16384	4	random	100.0	100.0	87.8	181.1	4.4	4.8
vLLM	16384	4	shared_prefi	100.0	100.0	44.4	59.8	3.6	4.0
vLLM	16384	8	random	100.0	100.0	117.3	371.6	5.7	6.2
vLLM	16384	8	shared_prefi	100.0	100.0	63.2	84.3	3.8	4.3
vLLM	16384	16	random	100.0	20.0	674.0	953.4	8.4	9.4
vLLM	16384	16	shared_prefi	100.0	100.0	88.7	107.0	4.4	4.9
vLLM	16384	32	random	100.0	60.0	425.7	710.6	10.8	11.4
vLLM	16384	32	shared_prefi	100.0	100.0	118.5	169.1	4.9	5.4
vLLM	16384	64	random	100.0	55.0	462.0	753.3	10.6	11.3
vLLM	16384	64	shared_prefi	100.0	100.0	105.4	124.0	4.9	5.4
vLLM	16384	128	random	100.0	55.0	451.4	770.5	10.6	11.7
vLLM	16384	128	shared_prefi	100.0	100.0	117.9	137.0	4.9	5.3
vLLM	32768	1	random	100.0	100.0	129.1	139.4	5.1	5.5
vLLM	32768	1	shared_prefi	100.0	100.0	31.4	35.2	3.3	3.7
vLLM	32768	2	random	100.0	100.0	122.5	167.3	4.3	4.6
vLLM	32768	2	shared_prefi	100.0	100.0	36.7	46.6	3.4	3.9
vLLM	32768	4	random	100.0	100.0	133.1	365.0	5.5	6.3
vLLM	32768	4	shared_prefi	100.0	100.0	44.5	55.0	3.5	4.0
vLLM	32768	8	random	100.0	65.0	144.5	884.6	8.3	9.2
vLLM	32768	8	shared_prefi	100.0	100.0	63.1	80.5	3.8	4.3
vLLM	32768	16	random	100.0	20.0	1357.6	1895.7	13.4	14.6

Engine	Ctx	C	Workload	Ok%	Goodput%	TTFT p50	TTFT p99	TPOT p50	TPOT p99
vLLM	32768	16	shared_prefi	100.0	100.0	95.5	116.8	4.3	4.8
vLLM	32768	32	random	100.0	0.0	1417.5	2016.7	16.5	18.3
vLLM	32768	32	shared_prefi	100.0	100.0	125.0	155.1	4.7	5.2
vLLM	32768	64	random	100.0	0.0	1387.1	1933.5	16.8	18.6
vLLM	32768	64	shared_prefi	100.0	100.0	145.7	180.1	4.7	5.2
vLLM	32768	128	random	100.0	0.0	1513.7	2090.8	16.6	18.7
vLLM	32768	128	shared_prefi	100.0	100.0	126.5	157.1	4.7	5.2